AKSHITA JAIN

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Data Analyst at the intersection of healthcare and AI, working across analytics, automation, and LLM-based solutions within a healthcare technology environment. Hands-on with Power BI, LangChain, LangFlow, GPT-4o, RAG workflows, and agentic pipelines, bringing together data expertise and generative AI to build intelligent healthcare solutions

EXPERIENCE

KareXpert — **Sep 2025– Present**
Data Analyst & AI — *Delhi, India*

- Developing a **RAG-based system** for intelligent retrieval from large volumes of clinical and operational healthcare records, designing document ingestion pipelines, optimizing chunking strategies, and fine-tuning embedding models to improve retrieval precision across multi-turn clinical queries
- Iterating on **prompt engineering and vector store integration** to ensure the system surfaces the most contextually relevant clinical information before passing it to the LLM, continuously refining retrieval accuracy and reducing hallucinations based on real query testing
- Building an **NLP pipeline for medical transcription**, converting doctor-patient voice interactions into structured clinical notes using ASR and language models, with post-processing steps for named entity recognition of medical terms, diagnoses, and ICD-10 code extraction
- Designing the document structuring layer that transforms raw transcribed text into EHR-compatible formats, processing 100+ clinical encounters daily, ensuring notes are accurate, compliant, and ready for downstream consumption by hospital information systems.

1Gen — **Sep 2022 – Jan 2023**
Data Science Intern — *Delhi, India*

- Leveraged **Python, SQL, and machine learning algorithms (regression models, clustering techniques**) to analyze **15,000+ EHRs**, uncovering significant trends that boosted precision medicine adoption by **20% and reduced patient stress by 15%.**
- Enhanced **operational efficiency by 30%** by automating EHR **data processing with Python**, reducing data handling time, and advancing mindfulness-based healthcare insights through Tableau dashboards.

PROJECTS

StartupAI - AI Powered Startup Blueprint Generator Blog — **May 2026**
Technologies: *Python, FastAPI, React, Groq (Llama 3.3-70B), Supabase, Vercel, Render* — Startup AI

- Built **StartupAI,** a **7-agent AI pipeline** that takes a startup domain as input and generates a complete business blueprint covering market research, competitor analysis, revenue models, MVP planning, and an investor pitch deck in one fully automated run
- Designed a **sequential multi-agent system** where each of the 7 agents passes its output as context to the next, ensuring every section builds on the previous one across Market Research, Competitor Analysis, Revenue Model, MVP Planner, Synthesis, Pitch Generator and PPTX Export
- Deployed the full app on **Vercel and Render** with **Google OAuth** login via Supabase, run history, and PDF and PPTX download support, with **26 real users** onboarded organically within weeks of launch.

ReflexRAG – Clinical PDF Research Assistant — **Apr 2026**
Technologies: *Python, FastAPI, Next.js, OpenAI GPT-4o mini, FAISS, BM25, AWS EC2, nginx, Netlify* — *Reflex RAG*

- Built a **9-stage RAG pipeline** for clinical PDFs combining **FAISS + BM25** hybrid retrieval with **RRF fusion, MMR diversity filtering, and cross-encoder reranking** across up **to 200+ document** chunks per query
- Implemented **Self-Reflection RAG** with **2-iteration grading loops** that verify retrieval relevance and answer faithfulness, reducing hallucinations on domain-specific clinical tables with rule-based table parsing
- Deployed FastAPI backend on **AWS EC2 t3.micro** with nginx reverse proxy, Let's Encrypt SSL, and DuckDNS domain; Next.js frontend on Netlify achieving 24/7 uptime at $0 infrastructure cost

GeneSight- Genomic Intelligence Pipeline — **Feb 2026**
Technologies**:** *Python, LangChain, GPT-4o, Neo4j, Neon PostgreSQL, Clerk, Vercel, Render* — *Gene Sight*

- Built **GeneSight**, **a 9-agent AI pipeline** for genomic research that autonomously generates and validates novel gene-disease hypotheses, spanning task planning, literature mining across **60 real PubMed papers** via NCBI Entrez API, knowledge graph construction in Neo4j, and hypothesis generation using **GPT-4o**, all completing in under **90 seconds**
- Engineered a concurrent hypothesis validation system where a Validator agent cross-references **5 hypotheses against 60 papers** simultaneously using GPT-4o-mini, calculating confidence scores based on real paper evidence, supported by a rule-based Evaluator agent **achieving 100% hallucination-free** quality checks with no LLM dependency
- Deployed a production-ready system with **Clerk authentication**, usage tracking via **Neon PostgreSQL**, and full-stack deployment on **Vercel and Render,** integrating 9 specialized agents, cross-referencing genomics data across NCBI and UniProt, and generating structured reports in plain English

SKILLS

Languages: Python, R, SQL
Generative AI & LLMs: LangChain, LangFlow, RAG (Retrieval-Augmented Generation), Agentic Workflows, Vector Databases, Embeddings, Fine-Tuning(LoRA, PEFT), HuggingFace Transformer, OpenAI API
Frameworks/Libraries: FastAPI, PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy
Database: MySQL, MongoDB, FAISS, Neo4j
Tools/Visualisation: Tableau, PowerBI, Git, GitLab

EDUCATION

Boston University — **Sep 2023 – Jan 2025**
Master of Science in Applied Data Analytics — *Boston, MA*

Guru Gobind Singh Indraprastha University — **Sep 2019 – Jul 2023**
Bachelor of Technology in Computer Science & Engineering — *Delhi, India*